

Immersive Visualization for Surgical Planning Through Mental Model Elicitation: Work in Progress

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ABSTRACT

Extended Reality (XR) technologies have opened new possibilities for image visualization in healthcare, but eliciting user requirements for XR development is challenging. Human-computer interaction can help obtain User Mental Models, mostly unexplored in XR for health. An ongoing project aims to address this gap by studying the skills and backgrounds of radiologists and surgeons using the same XR tool in Preoperative Surgical Planning for Complex Malformations through elucidating mental model, performing technical review and proposing Potential Interaction Models or *Interaction Concepts*. The work aims to contribute insights for designing more intuitive, immersive medical visualization tools.

Index Terms: Human-centered computing—Visualization—Visualization techniques—Treemaps; Human-centered computing—Visualization—Visualization design and evaluation methods

1 INTRODUCTION

Extended Reality (XR) technologies are complementing healthcare procedures in medicine. However, the usability and performance of clinicians using XR medical technologies need improvement in terms of interaction as there is a lack of information on how to address the wide context of use in healthcare [2]. This work in progress aims to provide a design and evaluation contribution for XR in medical image visualization from a user-centered perspective, applying an iterative process that involves Mental Model elicitation, technical review and the proposition of Interaction Concepts (IC) that match both surgeon and radiologist requirements in the context of preoperative surgical planning of complex malformations.

Stakeholder	Task	PC	VR module	VR-only
Radiologist	3D navigation	✓	✓	✓
	3D 30° navigation	✓	✓	✓
	Navigation (3D BoneScanner)	✓	✓	✓
	Craniotomy 3D/2D	✓	✓	✓
Surgeon	Navigation (2D and 3D)	✓	✓	✓
	Isolation	✓	✓	✓
	Dissect	✓	✓	✓
	Staple	✓	✓	✓
Both specialties	Assessment	✓	✓	✓
	Dynamic visualization	✓	✓	✓
	Assessment elements	✓	✓	✓
	Assessment elements	✓	✓	✓

Figure 1: Technical review of different acknowledged medical visualization software, four of them with VR features

2 BACKGROUND

Research on Mental Models has greatly contributed to the field of human-centered design (HCD). It enables us to gain a thorough understanding of how users think, perceive, and interact with products, services, and experiences. There are various methods to address the user mental model. In healthcare, medical augmented reality has been evaluated through user and clinical studies. However, the FDA has called for action to contribute to the evaluation of such technologies as a general evaluation framework cannot be used given the wide field of applications [2]. Design-centered approaches have been conducted, including Synthesis and Analytic approaches [1].

3 PROPOSAL APPROACH

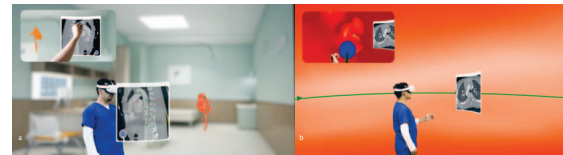


Figure 2: Endoscopy/Catheterism (E/C) interaction concept that mimics a popular medical image endoscopy feature. However, it is adapted to involve hand gestures and the simultaneous 2D and egocentric view for improving the user awareness

The methods used in the study included semi-structured interviews, direct observation, scientific and technical reviews, and concept design. There were three main stages: (1) eliciting the Mental Model for preoperative surgical planning, (2) identifying hardware and software that match the Mental Model, and (3) IC for the desired immersive interaction between radiologists and surgeons in the clinical context.

Two experienced health workers, a radiologist with 24 years of experience and a surgeon with 47 years of experience in general and cardiovascular surgery, were involved in the study and provided feedback. An external innovation expert in Medicine and Technological Innovation also performed external validation.

4 FINDING AND PRELIMINARY RESULTS

Figure 3 shows the resulting Mental Model that considers background, performance and capabilities. The observational process was performed with physicians who were exposed to an in-house XR technology system during three years. In general terms, the mental model varies according to each specialty. Therefore, A single system XR interaction system may affect the user performance.

Those respective mental models were adapted as Clinical Interaction Requirements (CIR) presented as follows:

Radiologist	Surgeon
Background and past experiences • Training in medical image visualization software using 2D inputs. • Exposed for three years to an in-house VR system for surgical planning	Background and past experiences • Training in operating patients based on clinical and medical imaging data. No expertise in medical image visualization software. • Exposed for three years to an in-house VR system for surgical planning
Performance • Prefers rectilinear movements for navigating the dataset to preserve orientation. • Experiences difficulties using classical immersive interaction techniques (E.g., Drag-and-drop), presenting misorientation.	Performance • Avoid rectilinear movements and interaction techniques close to classical medical visualization tools. • Appropriate classical immersive interaction techniques (E.g., Drag-and-drop) to move intuitively between structures but can feel misoriented easily.
Expectations • Desire to highlight key anatomy without performing complex manipulation tools/segmentation. • Prefers interaction tools with a degree of similarity to classical PC medical visualization software.	Expectations • Desire interaction techniques that mimic the surgery procedure while using 3D medical data. • Desire interacting with dynamic or 4D data to identify the behavior of structures in movement (e.g., Cardiac cycle).

Figure 3: The resulted mental model is addressed separately. Clinicians have different backgrounds about common experience using an in-house XR technology setup [3].

- 2D Navigation: the essential visualization task for patient diagnosis and treatment. It should be available no matter the software platform or clinical application.
- Multiplanar Reconstruction (MPR): Complementary visualization/navigation method to generate transverse planes according to the intended structure position and shape (rectilinear or curved).
- 3D Navigation provides a complementary perspective of the 3D Dataset (Render or Mesh), with many interaction possibilities. The Surgeon-Radiologist contexts should consider the following modalities:
 - Rectilinear mode for the Radiologist trained for linear (Slice-to Slice) screening.
 - Grab-and-Drop Navigation for the Surgeon constantly performing procedures and perceiving the anatomy directly from the patient.
- Manipulation: The capability of tag, annotate and establish Regions of Interest (ROI) for a deep understanding of intended structures. The Surgeon-Radiologist contexts should consider the following modalities according to their cognitive requirements: Annotations, 2D/3DROI, Incision, Dissection, and Suture. Dynamic Visualization and Registration are also expected. Finally, Awareness Elements are required to previous tools intuitively (Location, Orientation, Number of Tags, among others).

A technical review was performed to identifying which available commercial or open source systems match with the mental model's CIR 1. Therefore, for each CIR, an IC was presented and evaluated for the clinicians.(Figure 2). The concept was proposed as an illustrative video without making a functional prototypes to safe costs and efforts. A summary of the clinicians and external expert perceptual response is presented in Table 4.

5 FINAL REMARKS

Utilizing a Mental Model method is an effective approach for generating IC that can be utilized in preoperative surgical planning assisted by XR. The results obtained so far demonstrate the positive reception of two of the eight interaction concepts by healthcare professionals. Using a video format enables the identification of clinician perception on the IC without requiring the use of functional prototypes. This saving valuable time and effort that can be utilized for further research in the interaction design of immersive environments for complex malformations. A significant challenge is

IC	Radiologist			Surgeon			Innovation expert		
	Novelty	Acceptance	Adjustments	Novelty	Acceptance	Adjustments	Novelty	Acceptance	Adjustments
E/C		✓	✓		✓			✓	✓
T3DN		✓	✓	✓	✓	✓	✓	✓	✓
S&G	✓	✓	✓	✓	✓		✓	✓	✓
SROI	✓	✓	✓	✓	✓		✓	✓	✓
I2DN			✓						✓
IMPRN			✓					✓	✓
I3DN		✓	✓	✓	✓			✓	✓
IR								✓	✓

Figure 4: The IC Endoscopy/Catheterism (E/C), Through 3D Navigation (T3DN), Slice and Grab (S&G), Sculp ROI (SROI), Immersive 2D navigation (I2DN), Immersive MPR navigation (IMPRN), Immersive 3D navigation (I3DN), and Image Reference (IR).

finding users with the specific profile addressed in this study, namely those with extensive experience in healthcare and using XR. Further research is needed to iterate evaluation, design, and theorizing of a comprehensive mental model for the presented clinical context.

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